BuildMind

AI-Powered Residential House Design and Construction Planning System

Full Consolidated Chat & Project Handoff Document

Prepared from the BuildMind project discussions, decisions, dataset research, cost-estimation planning, AI-agent design, and Git knowledge-repository workflow.

Document purpose

This document is a structured project record, not a verbatim transcript. It is intended to let a new developer, team member, or future AI assistant understand the complete BuildMind direction and continue the work without losing context.

1. Executive Summary

BuildMind is an AI-assisted residential house design and construction planning platform. The system is intended to collect user requirements, generate valid residential floor-plan alternatives, render 2D plans, create a matching 3D model, estimate construction cost using verified Sri Lankan construction data, and support iterative design changes through an AI assistant.

A key architectural decision is that the AI language model will not directly invent engineering dimensions, structural quantities, construction prices, or legal compliance. Instead, the AI acts as an orchestrator: it understands the user, converts requests to structured inputs, calls deterministic engines, retrieves validated datasets, and explains the resulting design and cost information.

```
User Requirement
  -> AI Requirement Agent
  -> Structured Project JSON
  -> Rule / Constraint Engine
  -> Parametric Building Model
  -> 2D Drawing Engine
  -> 3D Model Engine
  -> BOQ / Cost Engine
  -> Validation Engine
  -> AI Explanation
```

2. Project Identity and Scope

Item	Current Decision
Project Name	BuildMind
Project Type	AI-Powered Residential House Design and Construction Planning System
Primary Domain	Residential house planning and preliminary construction estimation
MVP Focus	Single residential projects, mostly rectangular layouts, rule-driven 2D planning, basic 3D, preliminary cost estimation, AI assistance
Primary Market Context	Sri Lanka, with current cost-research emphasis on Northern Province / Jaffna-compatible data
Core Principle	AI interprets and orchestrates; deterministic engines calculate and validate

3. User, Admin, and AI Responsibilities

Role	Main Inputs / Control	Main Responsibilities	Important Limits
User	Land dimensions, location, road direction, north direction, rooms, parking, budget, style, preferences	Create projects; provide requirements; generate and compare plans; request changes; view 2D/3D; view estimates; save/export	Does not manually manage system reference datasets
Admin	Verified datasets, prices, labour rates, effective dates, source metadata	Manage users/projects; maintain room standards; adjacency; rules; construction cost data; price history; dataset versions; source verification	Must not silently overwrite history or activate unverified data

AI	Natural-language user requests + structured tool results	Understand requests; detect missing data; call tools; explain design decisions; guide plan refinement; explain cost output	Must not invent engineering values, current prices, legal approval, or structural safety claims

4. User Journey and Admin Journey

4.1 User Journey

Register / Login
-> Dashboard
-> Create New Project
-> Enter Land & Location Details
-> Enter Room Requirements
-> Enter Budget & Preferences
-> AI Requirement Processing
-> Missing Information Check
-> Constraint / Rule Validation
-> Generate Multiple Floor-Plan Alternatives
-> Score & Compare
-> 2D Plan View
-> Cost Estimate
-> User / AI Changes
-> Revalidation
-> Final Selection
-> 3D Visualization
-> Save / Export

4.2 Admin Journey

Admin Login
-> Dashboard
-> User / Project Management
-> Room Standards
-> Building Rules
-> Construction Cost Management
-> Labour Rates
-> Dataset Management
-> Architecture Styles
-> AI Configuration
-> Source Verification
-> Error / Report Monitoring

5. Seven-Member Dataset Research Split

Member	Dataset Area	Main Data to Collect	Expected Output
1	Weather / Climate	Temperature, humidity, rainfall, wind, sun/light, climate-region, ventilation factors	Climate dataset + sources
2	Construction Cost	Materials, labour, work items, rate analysis, productivity, reinforcement, formwork, transport, wastage, OH&P, VAT, district rates	Cost datasets for BOQ/cost engine

3	Room Dimensions / Space Standards	Bedroom, bathroom, kitchen, living, dining, corridor, parking, stairs, store/laundry minimum and preferred dimensions	Room standards dataset
4	Room Adjacency	Kitchen-dining, living-entrance, bedroom grouping, master-bath, parking-entrance, privacy, avoid/noise relationships	Adjacency / functional relationship dataset
5	Building Rules / Regulations	Setbacks, plot coverage, height, road boundary, ventilation, parking, drainage, local authority rules	Rules dataset
6	Door / Window / Furniture Dimensions	Door/window sizes and swing, bed/sofa/table/wardrobe/kitchen/toilet clearances	Object dimension dataset
7	Architecture Styles / Plan Templates	Modern, tropical, traditional Sri Lankan, minimalist, contemporary, courtyard, roof/facade/material preferences, sample plans	Style/template dataset

6. Current Technology Direction

Layer	Preferred Technology	Purpose
Frontend	Angular	User/admin UI, project forms, plan editor/viewer
2D Plan	Konva.js / SVG / Canvas	Dimensionally controlled plan rendering
3D	Three.js	Browser-based parametric 3D visualization and walkthrough
Backend	Python FastAPI	AI orchestration, APIs, geometry/cost services
Database	PostgreSQL	Projects, datasets, rates, versions, history
AI	Pretrained LLM + RAG + tool calling	Natural-language understanding and orchestration
Optional advanced tools	Blender / MCP / optimization libraries	Asset generation, rendering experiments, advanced workflows

7. Floor-Plan Generation Architecture

User Requirements
-> Requirement Extraction
-> Structured JSON
-> Room Standards
-> Adjacency Rules
-> Orientation / Climate Rules
-> Constraint Solving
-> Candidate Layout Generation
-> Geometry Validation
-> Circulation / Opening Validation
-> Plan Scoring
-> Multiple Alternatives
-> 2D Rendering
-> 3D Conversion

A central design decision is that 2D, 3D, and cost calculation should share the same parametric geometry model. For example, if a wall is 4.0 m in the source geometry, the 2D drawing, 3D wall, and BOQ quantity should all derive from that same value.

```
{
  "room": "Bedroom",
  "x": 2.0,
  "y": 4.0,
  "width": 3.6,
  "length": 4.0,
  "wallHeight": 3.0,
  "doors": [],
  "windows": []
}
```

8. Engineering-Oriented Engines Required

Engine	Purpose	Example Output
Geometry Engine	Exact coordinates, dimensions, areas, wall/opening geometry	Parametric room/wall model
Constraint Engine	Room standards, setbacks, adjacency and other rules	Pass/fail constraint set
Circulation Validator	Check usable access paths and room reachability	Accessibility/circulation warnings
Opening Validator	Check door/window placement and conflicts	Door/window validation
Orientation Engine	Apply north/light/ventilation preferences	Orientation score
Layout Generator / Solver	Generate candidate arrangements	Multiple valid layouts
Scoring Engine	Rank candidate plans	Plan score and reasons
BOQ Engine	Convert geometry to measurable work quantities	m3, m2, kg, No., m
Cost Engine	Apply rates, coefficients and adjustments	LKR estimate breakdown
2D CAD-style Renderer	Render dimensionally consistent plans	SVG/Canvas plan
3D Parametric Renderer	Create walls/floors/doors/windows/roof from geometry	Three.js model
Validation Report	Explain checks and failures	Human-readable validation report

9. Construction Cost Estimation Architecture

The project discussion established that material price data alone is insufficient. BuildMind needs both a rate-analysis/resource-consumption layer and a dynamic price layer.

```
House Geometry
-> Measurement Engine
-> Work Item / BOQ Engine
-> Rate Analysis / Resource Coefficients
-> Material Quantities
-> Labour Quantities
-> Equipment Quantities
-> Current Price Database
-> District / Location Adjustment
-> Wastage / Transport
-> Overhead / Profit
-> VAT / Tax
-> Final Preliminary LKR Estimate
```

9.1 Example: Why resource coefficients are required

For a concrete or RCC work item, BuildMind must know the work quantity and the resources required per unit. The exact coefficients must come from validated engineering/rate-analysis sources; the AI must not guess them.

Work Item	Resource Type	Typical Resource Fields	Notes
Concrete	Materials	Cement, sand, aggregate, water	Measured primarily by volume (m3)
RCC	Concrete + reinforcement + formwork	Concrete, steel, binding wire, shuttering, labour	Reinforcement/formwork should not be hidden inside a single guessed number
Masonry	Materials + labour	Bricks/blocks, mortar, cement, sand, mason/helper	Often measured by area/volume depending on item
Plaster / Tiling / Painting	Finishing resources	Mortar/adhesive/grout/paint/ coats/labour	Measured by surface area
MEP	Items + length + labour	Pipes, fittings, sanitary fixtures, wires, sockets, DBs	Requires itemized work rules

10. Planned Construction Cost Datasets

Dataset File	Purpose
work_items.csv	Item code, category, description, unit, district/region, rate, effective date, source
measurement_rules.csv	Formula, required inputs, output unit, source/reference
resource_coefficients.csv	Resource required per unit work; material/labour/equipment quantities
material_prices.csv	Current/historical material prices, brand/specification, district, source
labour_rates.csv	Trade/skill level, day/hour unit, district rates, effective date
labour_productivity.csv	Output per labour-day or equivalent productivity norm
reinforcement_data.csv	Bar diameter, kg/m, spacing-based kg/m2, source
formwork_data.csv	Element/formwork areas and resource/labour factors
masonry_data.csv	Block/brick size, mortar and labour relationships
finishing_data.csv	Plaster, tiles, paint, ceiling resource relationships
plumbing_data.csv	Pipe/fitting/fixture work items and rates
electrical_data.csv	Electrical points/items, materials and labour
equipment_rates.csv	Mixer, vibrator, excavator, scaffolding/tools rates
transport_rates.csv	Delivery / distance / regional transport adjustments
adjustment_rules.csv	Wastage, overhead, profit, VAT, transport inclusion, contingency
source_metadata.csv	Document, year, region, page, URL, verification state

11. Collected and Identified Construction Data Sources

The chat research identified and/or uploaded a broad set of official and reference documents. The most important sources for the current cost-estimation phase are listed below.

Source Group	Examples Collected	Primary Use	Priority
Northern Province SOR 2026	Building, Road, Irrigation, Material Testing, amendments, second-half updates	Current provincial work rates, labour rates, reinforcement data, OH&P/VAT/transport rules, productivity norms	**Very High**
Department of Buildings BSR	BSR 2020, 2021, 2023, 2024	Building work categories and schedule rates: excavation,	**Very High**

		concrete, masonry, roofing, MEP, finishing, glazing, painting, drainage, external works	
Central Province SOR	2025/2026 schedule documents	Cross-checking rates, specifications, market-survey methodology	**High**
Department of Census & Statistics	Selected Retail and Producer Prices 2021-2024	Historical material price trends and retail price context	**High**
CIDA / ICTAD references	Building specifications, CESMM / SMM, costing references	Measurement rules, specifications, rate-analysis methodology	**Very High**
Civil engineering handbooks	Civil Engineering Handbook; Civil Engineering Calculations; Civil Engineering Formulas	Calculation concepts and engineering reference logic	**Medium**
Other standards / manuals	Earthquake/seismic documents, drainage manuals	Specialized validation/reference; not primary Sri Lankan cost datasets	**Low / Secondary**

12. Important Findings Already Extracted

- Northern Province schedules provide district-sensitive construction context and are especially relevant to the Jaffna/Northern Province MVP.
- Labour rate tables include skilled, semi-skilled and unskilled categories, with district-specific values in amendments.
- Northern Province general data includes reinforcement bar weights and spacing-related unit weights.
- Official schedules include rules or notes for overhead, profit, VAT, loading/unloading, material wastage, transport, and special island transport considerations.
- Irrigation schedules include productivity-style norms expressed in labour-days or machinery output for certain work items, showing that resource/productivity datasets can be sourced from official schedules.
- Department of Buildings BSR documents cover major residential construction categories including excavation, concrete, masonry, roof work, plumbing/sanitary, electrical, finishing/flooring, glazing, painting, drainage and external work.
- DCS price bulletins provide historical price information for selected building materials, which is useful for price history and validation but should not be treated as a live current-price feed.

13. Dynamic Pricing and Admin Cost Management

Construction prices are dynamic. Therefore, BuildMind should not permanently hard-code material and labour prices. The admin module should maintain versioned, effective-dated, region-specific price records.

Admin Capability	Purpose
Add / edit price items	Maintain current supplier/material/labour data
District-specific pricing	Support Jaffna, Kilinochchi, Mullaitivu, Mannar, Vavuniya and future regions
Effective dates	Select correct rate for the required period
Active / inactive status	Prevent outdated rates from being used
Price history	Retain old rates instead of overwriting
Source metadata	Record supplier/official source and verification evidence
VAT / transport flags	Avoid double-counting or missing included costs

Import / export	Support CSV/XLSX-based research workflows

14. AI Strategy: Training, RAG, and Tool Calling

A major clarification in the discussion was that BuildMind does not need to train a foundation model from scratch. The recommended approach is to use a pretrained LLM and connect it to BuildMind knowledge, rules, and tools.

Approach	Use in BuildMind	Decision
Prompting / System Instructions	Define BuildMind behaviour and safety boundaries	Required
RAG / Retrieval	Retrieve room standards, rules, cost data and source references at request time	Very Important
Tool Calling	Call layout, validation, BOQ, cost and 3D services	Core architecture
Fine-tuning	Improve consistent requirement parsing or response style after enough labelled examples exist	Optional later
Training from scratch	Create a new LLM from zero	Not required / not recommended for this project
AI Brain = Existing LLM Knowledge = BuildMind datasets / RAG Memory = PostgreSQL project data Rules = Deterministic engines Actions = Backend tools / APIs 2D = Parametric geometry renderer 3D = Three.js Cost = BOQ + rate engine		

14.1 What should NOT be fine-tuned into the model

- Current cement/material prices
- Labour prices
- BSR/SOR rates
- Building rules that may change
- Room standards that require source/version traceability

These should remain in structured databases/retrieval layers so that updates can be applied without retraining the model.

15. Proposed BuildMind AI Agent Tools

Tool / Function	Responsibility
parse_requirements()	User message -> structured project requirements
validate_requirements()	Find missing or invalid inputs
get_room_standards()	Retrieve verified room-size rules
get_adjacency_rules()	Retrieve functional relationship constraints
get_building_rules()	Retrieve applicable building rules and source metadata
generate_layout()	Generate candidate geometry through solver/engine
validate_layout()	Apply dimensions, adjacency, circulation and opening checks
score_layout()	Rank alternatives and explain scores

render_2d()	Produce exact SVG/Canvas plan
generate_3d()	Create Three.js scene from parametric geometry
calculate_boq()	Convert geometry to measurable quantities
get_current_prices()	Retrieve active district/effective-date prices
estimate_cost()	Apply coefficients, rates and adjustments
modify_layout()	Apply user change request and regenerate
explain_plan()	Explain design choices, warnings and cost breakdown

16. Accuracy Goal and Engineering-Like Output

The project goal is not to generate decorative or dummy 2D/3D images. The output should be dimensionally consistent, rule-checked and traceable. However, a blanket 99% engineering accuracy guarantee should not be claimed without a defined benchmark and professional validation.

```
Dataset Quality
+ Correct Rules
+ Deterministic Formulas
+ Constraint Solving
+ Validation
+ Engineer-reviewed Test Cases
= High-quality measurable output
```

16.1 Recommended evaluation metrics

- Dimensional accuracy
- Room-rule compliance percentage
- Adjacency/circulation validity
- 2D-to-3D geometry consistency
- Quantity error percentage
- Cost-estimate deviation against reviewed benchmark
- Percentage of generated layouts that pass all configured rules

A future test suite should include standardized plots (for example 20x40, 30x50, 40x60 and other realistic cases) reviewed by a qualified civil/architectural professional. This is the correct way to make an accuracy claim meaningful.

17. AI Usage and User Capacity Planning

The discussion also clarified that monthly AI capacity is not normally a fixed "number of users". Practical capacity depends on API rate limits, token usage, request patterns, and monthly budget. For the university MVP, approximately 100 active users per month is a realistic target if AI requests are controlled and heavy geometry/cost work is performed by deterministic backend services rather than by the LLM.

- Use AI mainly for requirement parsing, tool orchestration and explanation.
- Keep 2D, 3D, BOQ and cost calculations in backend engines to reduce AI cost and improve consistency.
- Apply reasonable per-user limits and log usage during the MVP.
- Avoid resending unnecessary full conversation history; use structured project state and summaries.

18. BuildMind Git Knowledge Repository

To preserve continuity across different ChatGPT chats, team members and computers, a dedicated Git repository was created:

```
https://github.com/Kogulanathan69/BuildMind-Knowledge-.git
```

The local clone was successfully created under the user's OneDrive Desktop, and the first knowledge commits were pushed to the main branch.

Core File	Purpose
README.md	High-level repository and project overview
START-HERE.md	First-read context for future AI/developers
PROJECT-HANDOFF.md	Detailed architecture, scope and handoff information
CURRENT-STATUS.md	What is completed, in progress and missing now
PROJECT-DECISIONS.md	Finalized decisions and change-control history
NEXT-STEPS.md	Immediate and future task order

After major work:
1. Update CURRENT-STATUS.md
2. Update NEXT-STEPS.md
3. Update PROJECT-DECISIONS.md if a decision changed
4. Add/modify datasets or sources
5. git add .
6. git commit -m "docs: update BuildMind project progress"
7. git push

19. Recommended Repository Structure

```
BuildMind-Knowledge-/
  README.md
  START-HERE.md
  PROJECT-HANDOFF.md
  CURRENT-STATUS.md
  PROJECT-DECISIONS.md
  NEXT-STEPS.md
  docs/
    requirements/
    architecture/
    user-admin-ai/
    cost-estimation/
    research/
  datasets/
    01-weather/
    02-construction-cost/
    03-room-dimensions/
    04-room-adjacency/
    05-building-rules/
    06-object-dimensions/
    07-architecture-styles/
  sources/
    northern-province-sor/
    sri-lanka-bsr/
    dcs-prices/
    civil-reference/
  prompts/
  chat-history/
```

handoff/

20. Current Project Status

Area	Status
Project identity / scope	Defined
User / Admin / AI roles	Defined
7-member dataset split	Defined
Technology direction	Defined
2D/3D shared-geometry concept	Defined
Cost-estimation architecture	Defined
Dynamic pricing strategy	Defined
Official cost source collection	Strong base collected
Git knowledge handoff repository	Created and pushed
Structured construction datasets	Not yet fully extracted
Rate-analysis/resource coefficients	Still incomplete
Layout solver / scoring engine	Design direction defined; implementation pending
Production AI agent integration	Not started; should follow dataset/tool foundation

21. Immediate Next Steps

1. Extract the Northern Province 2026 Building Schedule of Rates into structured work-item data.
2. Create work_items.csv with item codes, descriptions, units, district rates, effective dates and source metadata.
3. Create labour_rates.csv from Northern Province rates and amendments.
4. Create material_prices.csv from official schedules, DCS history and later verified supplier updates.
5. Create reinforcement_data.csv and adjustment_rules.csv.
6. Identify and extract rate-analysis/resource coefficients for the first core residential work items.
7. Create measurement_rules.csv linking geometry to BOQ quantities.
8. Design the Admin Cost Management database tables and history/version model.
9. Implement deterministic cost calculation API before connecting the AI agent.
10. Build the first Requirement Parsing Agent only after the underlying structured inputs are defined.
11. After stable datasets and tools exist, connect the pretrained LLM via RAG + tool calling.
12. Create engineer-reviewed test cases and validation metrics before making accuracy claims.

22. Core Residential Cost Work Items for MVP

Site clearing	Floor tiling
Excavation	Wall tiling
Earth filling	Roofing
PCC	Ceiling
RCC footing	Doors
RCC column	Windows
RCC beam	Painting
RCC slab	Plumbing
Reinforcement	Sanitary fixtures
Formwork	Electrical
Brick/block masonry	Drainage
Plastering	External works

Floor screed	

23. Safety, Accuracy, and Professional-Use Boundaries

- BuildMind outputs should be described as preliminary/conceptual unless professionally reviewed.
- Final structural design, reinforcement and safety-critical engineering require qualified professional validation.
- Final building-law compliance and permit approval require confirmation by the relevant authority/professional.
- Cost estimates are not contractor quotations and may change with market conditions, site conditions, specification, labour availability and project complexity.
- Every important engineering/cost number should be source-traceable and versioned.
- AI-generated explanations should reflect tool results rather than fabricate unsupported values.

24. Standard Prompt for Future ChatGPT / AI Handoff

```
Read START-HERE.md, PROJECT-HANDOFF.md, CURRENT-STATUS.md,
PROJECT-DECISIONS.md, and NEXT-STEPS.md first.
Continue the BuildMind project from the documented current state.
Preserve finalized decisions.
Use documented datasets and official sources.
Do not invent engineering values or current prices.
Update CURRENT-STATUS.md and NEXT-STEPS.md after major progress.
Update PROJECT-DECISIONS.md whenever a new project decision is finalized.
```

25. Appendix A - Key Files and References Discussed / Uploaded

- Northern Province Schedule of Rates 2026 - Building
- Northern Province Schedule of Rates 2026 - Irrigation
- Northern Province Schedule of Rates 2026 - Road Works
- Northern Province Schedule of Rates 2026 - Material Testing
- Northern Province SOR amendments / second-half updates
- Department of Buildings - Construction BSR 2020
- Department of Buildings - Construction BSR 2021
- Department of Buildings - Construction BSR 2023
- Department of Buildings - Construction BSR 2024
- Central Province Approved Schedule of Rates 2025 / 2026
- Department of Census and Statistics - Bulletin of Selected Retail and Producer Prices 2021-2024
- Handbook of Civil Engineering Calculations
- Civil Engineering Formulas
- Civil Engineering Handbook 2016
- The Civil Engineering Handbook
- Additional seismic / drainage / academic reference documents (secondary to Sri Lankan official sources for this project)

26. Appendix B - Most Important Finalized Decisions

- BuildMind is the final project name.
- Angular is the main frontend direction.
- Python FastAPI is the preferred backend direction.
- PostgreSQL is the preferred database.
- Three.js is the preferred browser 3D engine.
- 2D, 3D and BOQ should derive from one parametric geometry source.
- AI should orchestrate tools, not replace engineering calculations.
- Construction prices are dynamic and must be admin-managed with history/effective dates.
- Sri Lankan official sources are the priority for cost/rules datasets.
- Material prices alone are insufficient; rate-analysis/resource coefficients are required.
- RCC cost must include concrete, reinforcement, formwork and relevant labour separately.
- Knowledge continuity is maintained through the BuildMind-Knowledge- Git repository.
- Fine-tuning is optional later; RAG + tool calling is the recommended first AI architecture.
- Accuracy should be measured with engineer-reviewed test cases rather than claimed without evidence.